

Right Node Raising and Flexible Cyclic Linearization

Aidan Malanoski

1. Introduction

Right node raising (RNR) is a phenomenon in which a string—the **pivot**—appears to be interpreted both in its pronounced position and in a position(s) where it is not pronounced—the **gap(s)**. The pivot follows the gap, hence the name *right* node raising. An example of right node raising is provided in (1). In this and following examples, the gap is indicated by underscoring, while the pivot is underlined.

- (1) Darius found ___ and Jasmine took the book.

(1) is a canonical example of right node raising, involving coordination with two conjuncts. However, right node raising may also occur across more than two conjuncts, as in (2), and may occur outside of coordination (Hudson 1976), as in (3). For expository purposes, when discussing right node raising in abstract terms, I will generally ignore these latter forms of right node raising. Unless otherwise noted, any point made about canonical right node raising also applies to less canonical cases.

- (2) Mahmoud lost ___, Darius found ___ and Jasmine took the book.

- (3) Those who ignore ___ dislike those who obey authority.

In what follows, I propose a theory of how right node raising constructions, and parallel structures more generally, are linearized. Section 1.1 introduces my theoretical assumptions about the nature of right node raising and linearization, and introduces the framework of Cyclic Linearization (Fox & Pesetsky 2005a). Section 2 investigates how right node raising constructions are linearized. As formulated in Fox & Pesetsky (2005a), Cyclic Linearization cannot linearize multidominant right node raising constructions. This motivates a modification of Cyclic Linearization, dubbed **Flexible Cyclic Linearization**, which broadly speaking allows linear order to be modified after linearization has taken place, but only as necessary to create a valid linear ordering. Flexible Cyclic Linearization straightforwardly linearizes multidominant right node raising constructions and derives many of their properties. Section 3 concludes the paper.

1.1. Background

1.1.1. Right node raising

There are three major theoretical approaches to right node raising. First is the **movement analysis** (e.g., Ross 1967, Sabbagh 2007), according to which the shared material undergoes **across-the-board (ATB) movement** to a position to the right of the coordinate structure (4). Second is the **(backwards) ellipsis analysis** (e.g., Wexler & Culicover 1980, Kayne 1994). According to this analysis, the conjuncts contain distinct but identical (or at least truth-conditionally equivalent) material, and the material in the non-final conjunct is deleted under identity with the material in the final conjunct (5). The third major approach is the **multidominance analysis** (e.g., McCawley 1982, Bachrach & Katzir 2017), where the pivot is syntactically present in every conjunct but pronounced in the final conjunct (6). In this example, I

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use coindexation to indicate multidominance. However, in subsequent sections of the paper, it should be assumed that repeated words in bracketed representations of syntactic structure represent a single multiply dominated constituent, unless otherwise noted.

- (4) Across-the-board movement analysis of right node raising
[CP [&P [TP Darius found t_i] [&' [TP Jasmine took t_i]]] [the book] $_i$]
- (5) Ellipsis analysis of right node raising
[CP [&P [TP Darius found ~~the book~~] [&' [TP Jasmine took the book]]]]]
- (6) Multidominance analysis of right node raising
[CP [&P [TP Darius found [the book] $_i$] [&' [TP Jasmine took [the book] $_i$]]]]]

Each of these approaches faces setbacks. For example, RNR can affect entities that cannot otherwise undergo movement or ellipsis (Wilder 2008, Bachrach & Katzir 2017, Larson 2018). Additionally, RNR does not always require strict morphosyntactic identity between the gap and pivot, contra the predictions of the multidominance analysis (Barros & Vicente 2011a). These are only some of the criticisms levelled against these approaches; for further discussion, I refer the reader to Bachrach & Katzir (2017) and Larson (2018).

In light of the challenges to these approaches, Barros & Vicente (2011b,a) propose an **eclectic analysis** of right node raising, according to which right node raising arises through either ellipsis or multidominance (at least; they acknowledge that additional mechanisms may be involved). As evidence, they provide diagnostics of ellipsis (namely, morphological mismatch and vehicle change) and multidominance (internal readings of relational adjectives, binomial *each*, and cumulative agreement)¹, and demonstrate that properties of ellipsis and properties of multidominance cannot be combined in a single instance of right node raising. Furthermore, Barros & Vicente (2011b: 34) hypothesize that multidominance is always available as a mechanism of producing right node raising, whereas (backwards) ellipsis is only available “if the corresponding type of forward ellipsis exists.” Chaves (2014) independently proposes a similar restriction on the role of ellipsis in right node raising.

Larson (2012) demonstrates that properties of ellipsis and multidominance can in fact be combined, apparently falsifying Barros & Vicente’s proposal. However, Belk et al. (forthcoming) argue that right node raising can simultaneously involve ellipsis and multidominance. Specifically, they note the logical possibility of an elided constituent in the initial conjunct containing material that is syntactically shared with the final conjunct, and provide prosodic evidence that such cases do indeed exist.

In conclusion, it currently appears that a grammatical analysis of right node raising is indeed possible, and that it will involve both multidominance and ellipsis. In what follows, I propose a theory of how multidominance generates right node raising, leaving a theory of ellipsis to future research.

1.2. Linearization

Linearization is the assignment of linear (temporal) order to a syntactic structure. Following Chomsky (1995), I assume that linearization occurs during the externalization of the syntactic structure. Consequently, linear order is absent from the narrow syntax.

Mathematically, we can model linear order as a two-place relation, that is, a set of ordered pairs. For concreteness, I will assume that this is a relation over **Vocabulary Items**, the phonological content inserted in terminal nodes.² We then define a linear ordering as in (7), with a working definition of precedence provided in (8). Note that this is not immediate precedence, so other Vocabulary Items may intervene between x and y .

¹ Binomial *each* is only used as a diagnostic in Barros & Vicente (2011b), whereas cumulative agreement is only used as a diagnostic in Barros & Vicente (2011a).

² Although I borrow the term *Vocabulary Item* from Distributed Morphology (Halle & Marantz 1993), I do not otherwise adopt that framework.

- (7) **(Linear) ordering:** Let Ω be the set of Vocabulary Items inserted in a sentence. A linear ordering L is any set $\{\langle x, y \rangle : x \in \Omega \text{ and } y \in \Omega \text{ and } x \text{ precedes } y\}$
- (8) **Precede:** Let Ω be the set of Vocabulary Items inserted in a sentence. Let $x, y \in \Omega$. Then x precedes y if the final phonetic segment in x is pronounced before the first phonetic segment in y .

Following Kayne (1994), it is generally accepted that linear orderings must be **transitive** (if $\langle x, y \rangle \in L$ and $\langle y, z \rangle \in L$, then $\langle x, z \rangle \in L$), **total** (or **connected**; if $x \neq y$, then $\langle x, y \rangle \in L$ or $\langle y, x \rangle \in L$), and **asymmetric** (if $\langle x, y \rangle \in L$, then $\langle y, x \rangle \notin L$).³ Asymmetry entails **irreflexivity** ($\langle x, x \rangle \notin L$). If a linear ordering is transitive, total, and asymmetric, then I refer to it as a **valid** linear ordering.

Now, suppose that we have two Vocabulary Items x, y and an ordering L such that x precedes y under L . We can then indicate precedence by membership in L : $\langle x, y \rangle \in L$. However, it is common in the literature (e.g., Fox & Pesetsky 2005a, Kusmer 2020) to indicate precedence using the binary operator $<$, so that instead of $\langle x, y \rangle \in L$, we write $x < y$ (or more precisely, $x <_L y$). Fox & Pesetsky (2005a) refer to statements of the form $x < y$ as **ordering statements**. Going forward, I will adopt this more familiar terminology and notation. However, two points are in order. First, we must be careful to distinguish *orderings*, which are two-place relations over Vocabulary Items, from *ordering statements*, which are statements of linear precedence (or equivalently, statements of membership in orderings). Second, I will sometimes refer to orderings informally as sets of ordering statements where this makes the exposition clearer.

1.3. Cyclic Linearization

Fox & Pesetsky's (2005a) **Cyclic Linearization** provides a theory of how linearization takes place. According to this proposal, transfer to the phonological interface happens in phases. In other words, upon completion of certain constituents (phases), Spell-Out (transfer to the phonological interface) takes place. When a phase is spelled out, it undergoes linearization. Under Cyclic Linearization, linearization obeys a property called **Order Preservation**, according to which "information about linearization, once established at the end of a given Spell-out domain, is never deleted in the course of a derivation" (Fox & Pesetsky 2005a: 6).

Cyclic Linearization is an alternative to Phase Theory (Chomsky 2000, 2001), and these theories differ in several key ways. Under Cyclic Linearization, the contents of a Spell-Out Domain are still available to the syntax after Spell-Out. Furthermore, there is no distinction between the phase and the Spell-Out Domain—the entire phase is spelled out. As such, there is no specific phase edge. Consequently, Cyclic Linearization must provide an alternative account of successive cyclic movement and island effects.

Fox & Pesetsky's account of these phenomena is rooted in Order Preservation. To see how Order Preservation accounts for successive cyclic movement and island effects, let us consider two alternate derivations of (9). To simplify exposition, I ignore head movement throughout this paper, unless otherwise noted. Before discussing this example, two points are in order. Following Chomsky (2000, 2001) and Fox & Pesetsky (2005b: 248, fn. 15), I take CP and vP to be phases. Additionally, because Cyclic Linearization is compatible with multiple algorithms for generating ordering statements, I will not adopt any specific such algorithm in this paper, instead taking it for granted that the linearization algorithm will generate usual English word order. The same point applies to the modification of Cyclic Linearization introduced in Section 2.2.

- (9) What did Mahmoud see?
- a. $[_{CP} \text{ what did Mahmoud } [_{vP} \text{ what Mahmoud see what}]]$
 - b. $*[_{CP} \text{ what did Mahmoud } [_{vP} \text{ Mahmoud see what}]]$

³ Kayne (1994: 4) refers to asymmetry as *antisymmetry*. However, as usually defined, antisymmetry differs from asymmetry in that asymmetric relations are necessarily irreflexive, but antisymmetric relations are not. Kayne (1994: 134) explicitly takes linear precedence to be irreflexive.

<i>what</i> < <i>Mahmoud</i>	<i>what</i> < <i>see</i>
	<i>Mahmoud</i> < <i>see</i>

Table 1: Ordering statements produced by Spell-Out of [_{VP} what Mahmoud see what] under Cyclic Linearization.

<i>what</i> < <i>did</i>	<i>what</i> < <i>Mahmoud</i>	<i>what</i> < <i>see</i>
	<i>did</i> < <i>Mahmoud</i>	<i>did</i> < <i>see</i>
		<i>Mahmoud</i> < <i>see</i>

Table 2: Ordering statements produced by Spell-Out of the CP in (9a). Ordering statements in bold are preserved from the earlier linearization of the vP.

Now, in (9a), *what* moves successive-cyclically to CP, with an intermediate position in Spec,vP. Spell-Out of the vP will produce the ordering statements in Table 1. Similar tables will be presented throughout this paper. The tables have no theoretical status; they are merely a convenient way of representing orderings.

Fox & Pesetsky crucially assume that only the most recent Merge of a constituent counts for the purposes of linearization. In other words, if a constituent moves, then linearization only takes into account its post-movement position. This assumption avoids reflexive ordering statements such as *what* < *what* and pairs of symmetric ordering statements such as *what* < *see* and *see* < *what*.

Spell-Out of CP will produce the ordering statements in Table 2. The result is a valid linear ordering which respects Order Preservation, and which generates the observed linear order of the sentence: *what* < *did* < *Mahmoud* < *see*.

Now, contrast this with the derivation in (9b), in which *what* does not move successive cyclically. Here, Spell-Out of vP generates the ordering in Table 3, which places *what* after *Mahmoud* and *see*. However, Spell-Out of CP generates the ordering in Table 4. Because *what* is linearized in its post-movement position, this ordering includes the ordering statements *what* < *Mahmoud* and *what* < *see*. However, these ordering statements contradict the ordering established by Spell-Out of the vP, violating Order Preservation. Thus, the non-successive-cyclic derivation in (9b) gives rise to an ordering contradiction, ruling that derivation out. This contrasts with the successive-cyclic derivation in (9a), which generated the observed linear order of (9) without giving rise to an ordering contradiction. In general, overt movement must be successive cyclic in order to be linearizable by Cyclic Linearization.

2. Linearizing right node raising

2.1. Right node raising under Cyclic Linearization

This section considers the problem that multidominant right node raising poses for Cyclic Linearization. Let us return to (1), repeated here.

- (1) Darius found ___ and Jasmine took the book.

When [_{VP} Jasmine took the book] is spelled out, linearization will generate the ordering statements in Table 5. Likewise, when [_{VP} Darius found the book] is spelled out, linearization will generate the ordering statements in Table 6. No issues arise at this stage; the ordering statements produced by Spell-Out of the vPs are completely compatible.

<i>Mahmoud</i> < <i>see</i>	<i>Mahmoud</i> < <i>what</i>
	<i>see</i> < <i>what</i>

Table 3: Ordering statements produced by Spell-Out of [_{VP} Mahmoud see what] under Cyclic Linearization.

<i>what < did</i>	<i>what < Mahmoud</i>	<i>what < see</i>	
	<i>did < Mahmoud</i>	<i>did < see</i>	
		<i>Mahmoud < see</i>	<i>Mahmoud < what</i>
			<i>see < what</i>

Table 4: Ordering statements produced by Spell-Out of the CP in (9b). Ordering statements in bold are preserved from the earlier linearization of the vP.

<i>Jasmine < took</i>	<i>Jasmine < the</i>	<i>Jasmine < book</i>
	<i>took < the</i>	<i>took < book</i>
		<i>the < book</i>

Table 5: Ordering statements produced by Spell-Out of [_{VP} Jasmine took the book].

<i>Darius < found</i>	<i>Darius < the</i>	<i>Darius < book</i>
	<i>found < the</i>	<i>found < book</i>
		<i>the < book</i>

Table 6: Ordering statements produced by Spell-Out of [_{VP} Darius found the book].

Table 7 presents the ordering generated by Spell-Out of the CP in (1); to save space, this table is presented in an abbreviated format. Here, check marks are used to indicate which ordering statements hold. For example, the checkmark at the intersection of the row labeled *Darius* and the column labeled *X < and* indicates that the ordering statement *Darius < and* has been generated. Two check marks indicates that an ordering statement was generated in a previous phase.

The ordering represented by Table 7 is not valid, for several reasons. First, it contains reflexive ordering statements such as *the < the*. Second, there are pairs of symmetric ordering statements such as *the < and* and *and < the*. Finally, ordering statements such as *book < the* contradict the ordering established in the vP phases. In short, the ordering relation represented by Table 7 violates Order Preservation, as well as the requirement that an ordering relation be asymmetric.

Recall that Fox & Pesetsky (2005a) stipulate that only the most recent Merge of a constituent counts when linearizing a constituent; this is how they linearize movement without creating incompatible ordering statements. In the present case, however, the two instances of [_{DP} the book] in (1) are parallel: there is no c-command relationship between them. Consequently, there is no way of determining which Merge of [_{DP} the book] happened first, and by extension, no way of determining where to linearize [_{DP} the book], other than by stipulation. Moreover, the present point is not limited to right node raising. In general, Fox & Pesetsky’s (2005a) formulation of Cyclic Linearization cannot linearize parallel structures, unless there is subsequent movement to a c-commanding position, as in across-the-board movement.

<i>X</i>	<i>X < found</i>	<i>X < and</i>	<i>X < Jasmine</i>	<i>X < took</i>	<i>X < the</i>	<i>X < book</i>
<i>Darius</i>	✓✓	✓	✓	✓	✓✓	✓✓
<i>found</i>		✓	✓	✓	✓✓	✓✓
<i>and</i>			✓	✓	✓	✓
<i>Jasmine</i>				✓✓	✓✓	✓✓
<i>took</i>					✓✓	✓✓
<i>the</i>		✓	✓	✓	✓	✓✓
<i>book</i>		✓	✓	✓	✓	✓

Table 7: Ordering statements produced by Spell-Out of the CP in (1) under Cyclic Linearization. Two check marks indicates that an ordering statement is preserved from the earlier linearization of the vPs.

2.2. Right node raising under Flexible Cyclic Linearization

The previous section demonstrates the problem that right node raising and other parallel structures pose for Cyclic Linearization: if a single constituent is merged in two positions that are not in a c-command relationship, then there is no way to determine which Merge happened first. Indeed, it is possible that the two Merge operations occurred simultaneously. Since there is no way of determining which Merge of a parallel-merged constituent happened first, there is no way of determining where to pronounce the constituent.

To resolve this, I propose a modification of Cyclic Linearization, which I call **Flexible Cyclic Linearization (FCL)**. At the center of FCL is the following proposal: rather than stipulating that only the most recent Merge counts during linearization, ordering statements may be deleted *in the phase in which they arise* as necessary to linearize a syntactic structure. This proposal maintains Order Preservation, which requires that ordering statements *established in previous phases* not be modified.

Several factors determine which ordering statements will be deleted. First, any ordering statements which violate Order Preservation (*i.e.*, which contradict ordering statements established when an earlier phase was spelled out) will be deleted. Second, reflexive ordering statements will be deleted. Third, one ordering statement out of each pair of symmetric ordering statements will be deleted in such a way that the resulting ordering is asymmetric, transitive, and total. Finally, economy will prevent ordering statements from being deleted unless they must be.

This proposal is akin to the system sketched in Fox & Pesetsky (2005b: 258), according to which “Ordering Statements are produced at every step of the derivation but [can] be deleted to resolve contradiction before the next Spell-Out takes place.” Ordering statements are initially placed in a “Provisional Ordering Table,” for which Order Preservation does not hold. Ordering statements are deleted as necessary to resolve contradiction, and at the end of the phase, the ordering statements in the Provisional Ordering Table are moved to the “Definitive Ordering Table,” for which Order Preservation does hold.

FCL differs from Fox & Pesetsky’s proposal in several ways. First, under FCL, ordering statements are not “produced at every step of the derivation.” Rather, ordering statements are only produced when Spell-Out of an entire phase occurs, as in the formulation of Cyclic Linearization in Fox & Pesetsky (2005a). However, under FCL, all positions in which a constituent appears are considered during linearization, producing a similar or identical effect to Fox & Pesetsky’s (2005b) proposal.

Second, Fox & Pesetsky (2005b: 258; emphasis added) say the following about their system.

In this more complex variant of our system, we could now imagine that the *syntax*, whenever it is faced with a choice among possible steps ... will make the choice that requires deletion of the fewest Ordering Statements from the Provisional Ordering table to resolve contradiction.

Apparently, Fox & Pesetsky envision a system in which the syntax can foresee ordering contradictions that will arise during Spell-Out. This introduces a look-ahead problem, rendering the proposal undesirable from the perspective of the Minimalist Program (Chomsky 2000). Under FCL, the syntax cannot predict the results of linearization. Consequently, the syntax cannot choose between derivational steps based on which will be easier to linearize. This reduces complexity in the syntax by erasing a look-ahead problem, but potentially increases complexity in the phonology, since the syntax may generate structures that are less easily linearized. However, this is not conceptually problematic, insofar as the syntax is primarily optimized for the semantic rather than phonological interface (Chomsky 2007).

Going forward, I will adopt Fox & Pesetsky’s notion of the Provisional and Definitive Ordering Table in slightly modified forms, introduced below. To synthesize the preceding discussion, the linearization of a phase proceeds as follows. When a phase is spelled out, ordering statements are generated, taking into account all positions in which a constituent appears. The set of ordering statements thus generated constitutes the **provisional ordering**. If the provisional ordering is not asymmetric, transitive, and total, then ordering statements are deleted as necessary to produce a valid ordering; I refer to this process as **pruning** (*cf.* Belk et al. forthcoming, whose proposal involves a different form of pruning). The pruning process obeys Order Preservation, leaving intact ordering statements established in previously linearized phases. If pruning successfully produces a valid ordering, this ordering is referred to as the **definitive ordering** for the phase.

<i>X</i>	<i>X</i> < <i>Darius</i>	<i>X</i> < <i>found</i>	<i>X</i> < <i>and</i>	<i>X</i> < <i>Jasmine</i>	<i>X</i> < <i>took</i>	<i>X</i> < <i>the</i>	<i>X</i> < <i>book</i>
<i>Darius</i>	✓	✓✓	✓	✓	✓	✓✓	✓✓
<i>found</i>			✓	✓	✓	✓✓	✓✓
<i>and</i>				✓	✓	✓	✓
<i>Jasmine</i>				✓	✓✓	✓✓	✓✓
<i>took</i>						✓✓	✓✓
<i>the</i>			✓	✓	✓	✓	✓✓
<i>book</i>			✓	✓	✓	✓	✓

Table 8: Provisional ordering for the CP in (1) under Flexible Cyclic Linearization. Two check marks indicates that an ordering statement is preserved from the earlier linearization of the vPs.

Let us return to the Spell-Out of the CP phase in (1). Under FCL, Spell-Out of CP will produce the ordering statements in Table 8. Table 8 differs from Table 7 in that the former includes *Darius* < *Darius* and *Jasmine* < *Jasmine*, since we are no longer assuming that only the most recent Merge counts for linearization.

As before, there are a number of problematic orderings. However, FCL allows us to resolve these orderings. First, we may delete the ordering statements in (10), which contradict previously established ordering statements. Strictly speaking, the problem is not that these ordering statements contradict previously established ones. Rather, the problem is symmetry. For example, we have both *Jasmine* < *the* and *the* < *Jasmine*. However, for the ordering statements in (10), their symmetric counterparts were established in a prior phase, and are therefore protected by Order Preservation. Consequently, there is no choice but to delete the ordering statements in (10).

- (10) Ordering statements in (1) that contradict previously established ordering statements
- a. *the* < *Jasmine*
 - b. *the* < *took*
 - c. *book* < *Jasmine*
 - d. *book* < *took*
 - e. *book* < *the*

We may also delete the ordering statements in (11), which are reflexive. Finally, we must delete one ordering statement from each pair of symmetric ordering statements listed in (12). Now, if we keep the ordering statements on the left, deleting *and* < *the* and *and* < *book*, then the resulting set of ordering statements will not be transitive. If we keep *the* < *and*, then we will have *the* < *and* and *and* < *took* but not *the* < *took*, which was deleted in (10). Note that we must keep *and* < *took*, or else *and* will not be ordered with respect to *took*. Likewise, if we keep *book* < *and*, then we will have *book* < *and* and *and* < *took* but not *book* < *took*, which was also deleted in (10). Consequently, we must delete the ordering statements *the* < *and* and *book* < *and*, instead keeping *and* < *the* and *and* < *book*.

- (11) Reflexive ordering statements in (1)
- a. *Darius* < *Darius*
 - b. *Jasmine* < *Jasmine*
 - c. *the* < *the*
 - d. *book* < *book*
- (12) Symmetric ordering statements in (1)
- a. *the* < *and* versus *and* < *the*
 - b. *book* < *and* versus *and* < *book*

<i>X</i>	<i>X < found</i>	<i>X < and</i>	<i>X < Jasmine</i>	<i>X < took</i>	<i>X < the</i>	<i>X < book</i>
<i>Darius</i>	✓✓	✓	✓	✓	✓✓	✓✓
<i>found</i>		✓	✓	✓	✓✓	✓✓
<i>and</i>			✓	✓	✓	✓
<i>Jasmine</i>				✓✓	✓✓	✓✓
<i>took</i>					✓✓	✓✓
<i>the</i>						✓✓

Table 9: Definitive ordering for the CP in (1) under FCL. Two check marks indicates that an ordering statement is preserved from the earlier linearization of the vPs.

Table 9 presents the ordering statements that remain after deleting the ordering statements discussed in the preceding paragraphs. If we look at the first ordering statement in each row, we can see that deleting the indicated ordering statements generates the observed linear order of the right node raising construction: *Darius < found < and < Jasmine < took < the < book*.

A number of properties of right node raising follow from this proposal. For example, right node raising (and left node raising) is subject to the so-called **Edge Restriction** (Bachrach & Katzir 2017: 3):

- (13)
- a.

Either [the shared material] α 's position is rightmost in all the nonrightmost constituents containing it, in which case it surfaces within the rightmost conjunct;
- b.

Or α 's position is leftmost in all the nonleftmost constituents containing it, in which case it surfaces within the leftmost constituent.

Clause (13a) requires that in right node raising, the gap must be final in the non-final conjunct; this property has been referred to as the “right edge condition” (Wilder 2008). Similarly, clause (13b) requires that in left node raising, the gap must be initial in the non-initial conjunct. Thus, (14) (= Bachrach & Katzir 2017: (4b)) is unacceptable because the gap is non-final in the initial conjunct.

- (14)
- * John gave __ presents and Bill kissed Sue.

Under FCL, the Edge Restriction is a consequence of Order Preservation: the pivot surfaces in the final conjunct because it is linearized at the “right edge” of—that is, after—all the material in the non-final conjuncts. If in a prior phase, the pivot is ordered before any material β in the initial conjunct, then this will give rise to an ordering contradiction in which the pivot is ordered before material in the initial conjunct (namely β) but is ordered after material in the final conjunct.

The FCL account also predicts the existence of **right node wrapping** (Whitman 2009), where the pivot is non-final in the final conjunct (15). In order to be realized in the final conjunct, Order Preservation requires that the pivot be ordered after all material in the initial conjunct. However, the pivot can be ordered before material in the final conjunct, as long as that ordering is respected in subsequent phases. This gives rise to right node wrapping.

- (15)
- Nkiru washed __ and Yngvarr put the dishes away.

FCL also correctly predicts that an RNR pivot need not be a constituent, as in (16) (based on Larson 2018: (9)). Under this approach, right node raising (or at least those cases that result from multidominance) is not a syntactic phenomenon, but rather an epiphenomenon of linearization. Given that it is not a syntactic phenomenon, we do not expect RNR pivots to be restricted to constituents.

- (16)
- I thought Becky's __ but you said Bruce's car exploded.

As discussed in Section 1, right node raising is not restricted to coordination (Hudson 1976); this fact is also predicted. Under FCL, right node raising is epiphenomenal, generated by the cyclic linearization

of parallel phases. Consequently, we predict right node raising to be possible not only in coordination, but wherever there is parallel structure.

Finally, FCL accounts for left node raising. Recall that RNR arises when shared material is ordered after other material in non-final phases. Order Preservation requires that ordering statements be preserved, leading to the pivot being spelled out in the final conjunct, where it follows the other material in both conjuncts. Now, imagine that shared material is ordered *before* other material in non-final phases. Order Preservation still requires that these ordering statements be preserved, so the pivot is spelled out in the initial conjunct, where it precedes the other material in both conjuncts. This gives rise to left node raising.

As the preceding discussion shows, FCL straightforwardly derives many properties of RNR—namely, those properties that are related to linear order. Unfortunately, space limitations prevent a full exposition of these facts, which will be presented more thoroughly in future work.

3. Conclusion

As formulated in Fox & Pesetsky (2005a), Cyclic Linearization cannot linearize parallel structures such as right node raising. To address this limitation, I proposed Flexible Cyclic Linearization, a modified version of Cyclic Linearization which allows for the deletion of ordering statements while abandoning the assumption that only the most recent Merge counts for linearization. Flexible Cyclic Linearization can linearize right node raising constructions, and predicts a number of their properties, including the existence of the Edge Restriction; the availability of right node wrapping; the fact that the pivot need not be a constituent; the availability of right node raising outside of coordination; and the existence of left node raising. That said, Flexible Cyclic Linearization does not account for all properties of multidominant right node raising, but this is unsurprising: Flexible Cyclic Linearization is a theory of linearization, so we do not expect it to derive properties (such as cumulative agreement; Grosz 2015) that are not obviously related to linearization.

In addition to linearizing multidominant structures, Flexible Cyclic Linearization provides a novel account of covert movement and distributed deletion (Fanselow & Ćavar 2002), as discussed during my oral presentation.⁴ Space limitations prevent me from exploring these consequences in this paper. For the present purposes, it will suffice to say that in addition to linearizing parallel structures, FCL provides novel accounts of variation in where a constituent is realized.

References

- Bachrach, Asaf & Roni Katzir. 2017. Linearizing structures. *Syntax* 20(1). 1–40.
- Barros, Matthew & Luis Vicente. 2011a. Right node raising requires both ellipsis and multidomination. *University of Pennsylvania Working Papers in Linguistics* 17(1). 1–9.
- Barros, Matthew & Luis Vicente. 2011b. The eclectic nature of Right Node Raising. Ms.
- Belk, Zoë, Ad Neeleman & Joy Philip. Forthcoming. What divides, and what unites, right-node raising. *Linguistic Inquiry*.
- Chaves, Rui P. 2014. On the disunity of right-node raising phenomena: extraposition, ellipsis, and deletion. *Language* 90(4). 834–886.
- Chomsky, Noam. 1995. Bare phrase structure. In Héctor Campos & Paula Kempchinsky (eds.), *Evolution and revolution in linguistic theory*, 51–109. Washington, D.C.: Georgetown University Press.
- Chomsky, Noam. 2000. Minimalist inquiries: the framework. In Roger Martin, David Michaels & Juan Uriagereka (eds.), *Step by step: essays on minimalist syntax in honor of Howard Lasnik*, 89–155. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2001. Derivation by phase. In *Ken Hale: a life in language*, 1–52. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2007. Approaching UG from below. In Uli Sauerland & Hans-Martin Gärtner (eds.), *Interfaces + recursion = language? Chomsky's minimalism and the view from syntax-semantics*, 1–29. Studies in Generative Grammar. Berlin/New York: Mouton de Gruyter.
- Fanselow, Gisbert & Damir Ćavar. 2002. Distributed deletion. In Artemis Alexiadou (ed.), *Theoretical approaches to universals*, 65–107. John Benjamins.

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- Fox, Danny & David Pesetsky. 2005a. Cyclic linearization of syntactic structure. *Theoretical Linguistics* 31. 1–45.
- Fox, Danny & David Pesetsky. 2005b. Cyclic Linearization and its interaction with other aspects of grammar: a reply. *Theoretical Linguistics* 31. 235–262.
- Grosz, Patrick Georg. 2015. Movement and agreement in right-node-raising constructions. *Syntax* 18(1). 1–38.
- Halle, Morris & Alec Marantz. 1993. Distributed Morphology and the pieces of inflection. In Kenneth Hale & Samuel Jay Keyser (eds.), *The view from building 20: essays in honor of Sylvain Bromberger*, 111–176. Cambridge, MA: MIT Press.
- Hudson, Richard A. 1976. Conjunction Reduction, Gapping, and Right-node Raising. *Language* 52(3). 535–562.
- Kayne, Richard. 1994. *The antisymmetry of syntax*. Cambridge, MA: MIT Press.
- Kusmer, Leland Paul. 2020. Optimal Linearization: word-order typology with violable constraints. *Syntax* 23(4). 313–346.
- Larson, Brooke. 2012. A dilemma with accounts of right node raising. *Linguistic Inquiry* 43(1). 143–150. Published under the name *Bradley Larson*.
- Larson, Brooke. 2018. Right node raising and nongrammaticality. *Studia Linguistica* 72(2). 214–260.
- McCawley, James D. 1982. Parentheticals and discontinuous constituent structure. *Linguistic Inquiry* 13(1). 91–106.
- Ross, John Robert. 1967. *Constraints on variables in syntax*. Cambridge, MA: Massachusetts Institute of Technology PhD dissertation.
- Sabbagh, Joseph. 2007. Ordering and linearizing rightward movement. *Natural Language & Linguistic Theory* 25(2). 349–401.
- Wexler, Kenneth & Peter W. Culicover. 1980. *Formal principles of language acquisition*. Cambridge, MA: MIT Press.
- Whitman, Neal. 2009. Right-node wrapping: multimodal categorial grammar and the ‘Friends in Low Places’ coordination. In Erhard W. Hinrichs & John Nerbonne (eds.), *Theory and evidence in semantics*, 235–256. Stanford, CA: Center for the Study of Language & Information.
- Wilder, Chris. 2008. Shared constituents and Linearization. In Kyle Johnson (ed.), *Topics in ellipsis*, 229–258. Cambridge, UK: Cambridge University Press.

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